Biomaterials Project:

Purpose

To apply knowledge of biomaterial properties to a material selection/ biomedical device design problem.

Suggested Process

1. **Select a Class III** biomedical device that has or is being developed for commercial use.

2. Identify the various major parts of the device or implant, provide a drawing or illustration and list the biomaterials used. This may involve contacting the manufacturer and/or locating the appropriate information within the patent literature. (Try www.patents.ibm.com and www.uspto.gov)

3. Justify the selection of the materials used with regards to similar successful applications, physical characteristics, suitability for processing/ manufacture, cost, and favorable federal regulations.

4. Provide an alternative biomaterial solution to at least one of the key components used in the device and justify your selection based upon relevant biological principals. List benefits and drawbacks of the existing components and those of the proposed alternative.

Project requirements

1. Project proposal due May 8th, half to one-page maximum. It should describe the selected device and propose ideas for completing the project. This will enable the instructor/TA to provide feedback on the project topic.

2. Oral project presentations of 13 minutes followed by 2 minutes for questions each on May 22nd.

Grading Policy

The project is worth 10% of your final grade, and includes evaluation of the preliminary report, final document and the oral project presentation.

Grading will favor reports that are complete, neat and provide reasonable rationale for the material selections.

Notes

Classification of medical devices is based on the duration of the device use, invasiveness and risk to the user.

Class I devices: crutches, bedpans, tongue depressors, adhesive bandages etc. These are short-term items, and do not internally contact the user.

Class II devices: hearing aids, blood pumps, catheters, contacts, electrodes etc. These are minorly invasive, but still external and relatively short-term. But the risk to user is greater.

Class III devices: cardiac pacemakers, intrauterine devices, intraocular lenses, heart valves, orthopedic implants, etc. These are long-term, considerably invasive and can pose immense risk to the user